

Contents

1 Chats

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1.1 You

Show me some non-basic examples how the Risch algorithm for integration works.

1.2 Gemini

Here's my plan to tackle that topic. Let me know if you need to make changes.

Risch Algorithm Integration Examples

Research Websites

- (1) Research the structural overview of the Risch algorithm, focusing on differential fields, tower of extensions, and the distinction between algebraic and transcendental extensions.
- (2) Find a detailed step-by-step derivation of the Risch algorithm applied to the function $f(x) = e^{x^2}$ to demonstrate how it proves the non-existence of an elementary anti-derivative.
- (3) Search for a worked example where the Risch algorithm successfully integrates a complex function involving both exponential and logarithmic terms (e.g., involving xe^x or $\ln(x)/x$).
- (4) Investigate the application of Hermite reduction and the Rothstein-Trager algorithm within the context of the Risch algorithm for handling rational parts and identifying logarithmic residues.
- (5) Find specific examples illustrating the solution of the Risch Differential Equation $y' + fy = g$ in the context of integrating exponential extensions.
- (6) Synthesize the findings to present distinct examples that highlight the algorithm's recursive nature and its method for reducing integration problems to algebraic differential equations.

More

Analyze Results

Create Report

Ready in a few mins

Edit plan Start research

1.3 You

Start research

1.4 Gemini

I've completed your research. Feel free to ask me follow-up questions or request changes.

Risch Algorithm Integration ExamplesCompletedJan 30, 10:49 AM
